Advanced Analysis and Classification of Modified MNIST Digit Dataset Using Dimensionality Reduction and Clustering

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Summary Report

Dataset Overview

- Utilized a large-scale MNIST variant with 50,000 images of handwritten digits (0-9).
- Each image is represented as a 28x28 pixel grid, flattened into input features.
- The dataset is balanced across digits (~10% per class), enabling use of accuracy as a primary metric.
- Due to large sample size and high dimensionality, efficient subsampling and dimensionality reduction are critical.

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a) Dimensionality Reduction and Data Exploration

1. **Subsampling Techniques:**
   - Employed *random subsampling* (5,000 samples) and *diversified sampling* (using fast diversified sampling algorithm) to capture representative data subsets.
   - Diversified sampling better preserves feature distribution spread, facilitating more reliable downstream analysis.
2. **Dimensionality Reduction Methods:**
   - Explored linear methods: PCA, Truncated SVD, Sparse PCA, and NMF.
   - Explored nonlinear methods: Kernel PCA (KPCA, with RBF kernel), t-SNE (with various perplexities), and UMAP.
   - Assessed through:
     - Visualization of components (2D and 3D scatter plots).
     - Scree plots depicting explained variance (or equivalent for nonlinear methods).
3. **Findings:**
   - PCA, Truncated SVD, and KPCA showed a similar decay in explained variance, motivating choosing ~60 components for modeling.
   - KPCA outperformed PCA and SVD in explained variance and cluster separation; Sparse PCA and NMF were less informative.
   - t-SNE and UMAP yielded superior visualizations with clear clustering of some classes; UMAP slightly better at separating overlapping classes (e.g., digits 6 and 9).

- Some classes (digits 2, 3, 8) exhibited diffuse clustering, indicating intrinsic difficulty and possible label noise.

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b) Classification Performance vs Training Size

1. **Sampling Strategy:**
   - Used *m out of n bootstrap* sampling (10 bootstrap sets of 1,000 samples each) for feasible and diverse model training.
2. **Dimensionality Reduction for Classification:**
   - Tested KPCA, Truncated SVD, and UMAP embedded in classification pipelines.
   - KPCA (60 components) achieved best validation accuracy (~0.83), followed by SVD (~0.81) and UMAP (~0.73).
   - Selected KPCA for final classification models.
3. **Classifiers:**
   - Trained six classifiers: KNN, QDA, Support Vector Machines (linear and kernel), Logistic Regression (kernelized), Random Forest (RF), and Neural Networks (NN).
   - Hyperparameters tuned using cross-validation and grid search; mostly stable except RF depth showing some variability.
4. **Results:**
   - Training set accuracy remained steady for all but KNN (which increased with training size).
   - Validation accuracy declined with smaller training size across models.
   - QDA degraded quickly, likely due to insufficient data to estimate Gaussian class distributions reliably.
   - RF, SVM, and NN remained more robust with moderate drops.

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c) Clustering Analysis and Class Recovery

1. **Methods Explored:**
   - *K-means clustering* on KPCA embeddings with varying k.
   - *DBSCAN* with extensive parameter tuning—found clustering poor due to density heterogeneity.
   - *Agglomerative hierarchical clustering* with multiple linkage criteria (single, complete, average, Ward).
2. **Findings:**
   - K-means showed no sharp elbow in WCSS plot but silhouette scores suggested ~8 clusters optimal.

- After label alignment, K-means clusters corresponded well with some digits (e.g., 0, 1), but merged several others (e.g., 4 & 7, 5 & 8, 6 & 9).
- DBSCAN largely failed to detect meaningful clusters for this dataset, often producing single large clusters or excessive noise points.
- Agglomerative clustering with Ward linkage performed better than other linkages; identified 8 clusters at suitable distance threshold.
- However, its adjusted clustering accuracy was lower than K-means, indicating sensitivity to feature space representation.

Conclusions and Insights

- Diversified sampling provides reliable representative subsets for exploring vast datasets.
- KPCA combines nonlinear mapping and variance preservation, yielding effective embeddings for classification.
- UMAP excels at visually uncovering complex cluster structure but may underperform slightly in direct classification tasks.
- Classifiers leveraging nonlinear embeddings (KPCA + SVM, RF, NN) offer best trade-off between accuracy and stability.
- Clustering partially recovers known class structure; however, overlapping digits and label noise limit perfect recovery.
- Future improvements could explore ensemble methods, refined embeddings, and active sample cleaning to handle mislabeled or ambiguous samples.

Key Interview Questions and Suggested Answers

1. **Q:** How did you handle the challenges posed by the large sample size and high dimensionality of MNIST?
   A: I applied sampling techniques like diversified and bootstrap sampling to reduce computational load while maintaining data representativeness, combined with dimensionality reduction (KPCA, UMAP) to reduce feature space complexity before modeling.
2. **Q:** Why choose KPCA over simpler methods like PCA for dimensionality reduction?
   A: KPCA captures nonlinear relationships better than PCA, producing embeddings that improved classification accuracy and cluster separation for complex image data with intricate variances not represented linearly.
3. **Q:** Which classifiers worked best with this dataset, and why?
   A: KPCA embeddings fed to SVM, Random Forest, and Neural Networks performed best, likely due to their ability to handle complex class boundaries and mitigate overfitting in high-dimensional spaces.
4. **Q:** How effective was clustering in recovering digit classes?
   A: K-means recovered major clusters well but merged several digit groups;

hierarchical clustering with Ward linkage had fewer clusters but more ambiguity. Density-based clustering was ineffective, suggesting cluster mixture and noise complexity.

5. **Q:** What are potential limitations experienced?
   A: Label noise and intrinsic similarity (e.g., between digits 6 & 9) complicated classification and clustering. Computational constraints limited exhaustive hyperparameter tuning and data size per training iteration.